

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I **SARSHINI.S 192321113** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

PROBLEM STATEMENT

A factory has three boxes containing colored balls.

- Box A contains 6 red, 4 blue, and 2 green balls.
- Box B contains 5 red, 7 blue, and 3 green balls.
- Box C contains 8 red, 2 blue, and 5 green balls.

One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue.

Find the probability that the selected box was Box C.

Let A, B, C be the events of choosing Box A, B, C.

Given: Box B is twice as likely as A and C

So,

- $P(A) = \frac{1}{4}$
- $P(B) = \frac{1}{2}$
- $P(C) = \frac{1}{4}$

Step 1: Probability of getting blue ball

- From Box A: $\frac{4}{12} = \frac{1}{3}$
- From Box B: $\frac{7}{15}$
- From Box C: $\frac{2}{15}$

Step 2: Total probability of blue ball

$$\begin{aligned} P(\text{Blue}) &= \frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15} \\ &= \frac{1}{12} + \frac{7}{30} + \frac{1}{30} \\ &= \frac{7}{20} \end{aligned}$$

Step 3: Apply Bayes Theorem

$$P(C | \text{Blue}) = \frac{P(C) \times P(\text{Blue} | C)}{P(\text{Blue})}$$

$$\begin{aligned}
 &= \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{7}{20}} \\
 &= \frac{2}{21}
 \end{aligned}$$

Final Answer:

$\frac{2}{21}$

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

BAYES' THEOREM PROBLEM

Question:

A parking facility has three sections:

- Section A: 5 red, 3 blue, 2 black cars (Total = 10)
- Section B: 4 red, 6 blue, 5 black cars (Total = 15)
- Section C: 7 red, 2 blue, 6 black cars (Total = 15)

A section is chosen randomly, but Section B is **twice as likely** to be chosen as Section A and Section C.

A car selected is **blue**.

Find the probability that the car came from **Section C**.

Let A, B, C be the events of choosing Section A, B, C.

Given: Section B is twice as likely

So,

- $P(A) = \frac{1}{4}$
- $P(B) = \frac{1}{2}$
- $P(C) = \frac{1}{4}$

Step 1: Probability of getting a blue car

- From Section A: $\frac{3}{10}$
- From Section B: $\frac{6}{15} = \frac{2}{5}$
- From Section C: $\frac{2}{15}$

Step 2: Total probability of blue car

$$\begin{aligned}
 P(\text{Blue}) &= \frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15} \\
 &= \frac{3}{40} + \frac{1}{5} + \frac{1}{30}
 \end{aligned}$$

Take LCM = 120:

$$= \frac{9}{120} + \frac{24}{120} + \frac{4}{120} = \frac{37}{120}$$

Step 3: Apply Bayes Theorem

$$\begin{aligned} P(C \mid Blue) &= \frac{P(C) \times P(Blue \mid C)}{P(Blue)} \\ &= \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{37}{120}} \\ &= \frac{2}{60} \div \frac{37}{120} = \frac{1}{30} \times \frac{120}{37} \\ &= \frac{4}{37} \end{aligned}$$

Final Answer:

$\frac{4}{37}$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand